

CSC-301 Introduction to Human Computer Interaction

CCP Project Report

Section: A



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MessHub – Digital Mess Management System

Project Overview

MessHub is a role-based mess management web application designed to manage daily attendance (Breakfast/Lunch/Dinner), tea consumption, billing periods, payments, and period reports. The primary modules of the system are as follow:

- Authentication & account recovery (login, forgot/reset password)
- Admin management: Members, Menu/Dish Plans, Mess Periods
- Daily operations: Attendance marking, Tea marking
- Member self-service: Attendance, Tea consumptions, Menus and Payments
- Finance & compliance: Payment tracking (including approval workflow), Report generation and export (CSV/PDF)

1. Requirements

1a) Identified Primary Users

Mess Admin (Role: Admin)

- Marks daily attendance for all members.
- Marks daily tea consumption entries for members.
- Manages members, dish plans (menu), and mess billing periods.
- Reviews/approves submitted payments.
- Generates reports for a billing period (CSV/PDF export).

Mess Member (Role: User)

- Views their attendance records.
- Verifies/declines each meal attendance entry to prevent disputes.
- Views tea records and verifies/declines entries.
- Views payments and remaining balance.

1b) Functional & Non-Functional Requirements

Functional Requirements

Authentication & Security

- User can log in using username/password.
- System blocks deactivated accounts and provides clear messages.
- User can request password reset via email link and reset password using a token.

Admin

- Admin can manage members (create/edit/deactivate)
- Admin can manage dish plans/menu (CRUD).
- Admin can manage mess periods (CRUD).
- Admin can mark attendance for all members for a selected date.
- Admin can assign meal-specific dish plans for the selected day-of-week.

- Admin can view attendance analytics via a period calendar heatmap.
- Admin can mark tea entries for all members and view period calendar.
- Admin can view payment summary for a period (charges vs paid vs remaining).
- Admin can approve/reject member-submitted payments (pending approvals).
- Admin can generate reports per period and export CSV/PDF.

Member

- Member can view his attendance and verify each meal.
- Member can view all the weekly dish plan menu.
- Member can decline incorrect attendance entries (per meal).
- Member can view his tea details and verify/decline entries or verify all.
- Member can submit his payment through stripe or other payment method (Approval Needed)
- Member can view his payments and see balance/charges (period-based).

Non-Functional Requirements

Usability: The system uses consistent navigation on all screens with a shared header and breadcrumbs. Clear feedback is provided for user actions through success messages, pending counts, and status badges, so users always know what is happening.

Performance: The application is designed to load pages quickly for normal mess sizes ranging from 30 to 200 members. Tables and calendar views remain smooth and easy to use without slowing down the system.

Security and Privacy: The system follows role-based authorization to ensure that only Admin users can access sensitive features. User passwords are securely stored using hashing, and password reset tokens are generated with expiry for added security.

Reliability: The system maintains data integrity by correctly managing period boundaries. Attendance, tea, and payment records are always summarized according to the selected or active period to avoid errors.

Accessibility: The interface supports keyboard navigation and provides visible focus indicators. Readable fonts and proper color contrast are used to meet basic WCAG accessibility standards and ensure inclusivity.

1c) User Needs, Pain Points, and Usage Scenarios

User Needs

The admin **needs** a **fast** and error-free way to mark **daily attendance** for multiple members. They also require a **clear** period **overview** to easily identify missing data and to manage finances by viewing dues, collection status, and **exporting records**. Members need transparency and trust in the system so they can verify or decline records. They also need a simple way to see **pending verifications** and **clearly** understand their meal, tea, and water **charges** for each period.

Pain Points

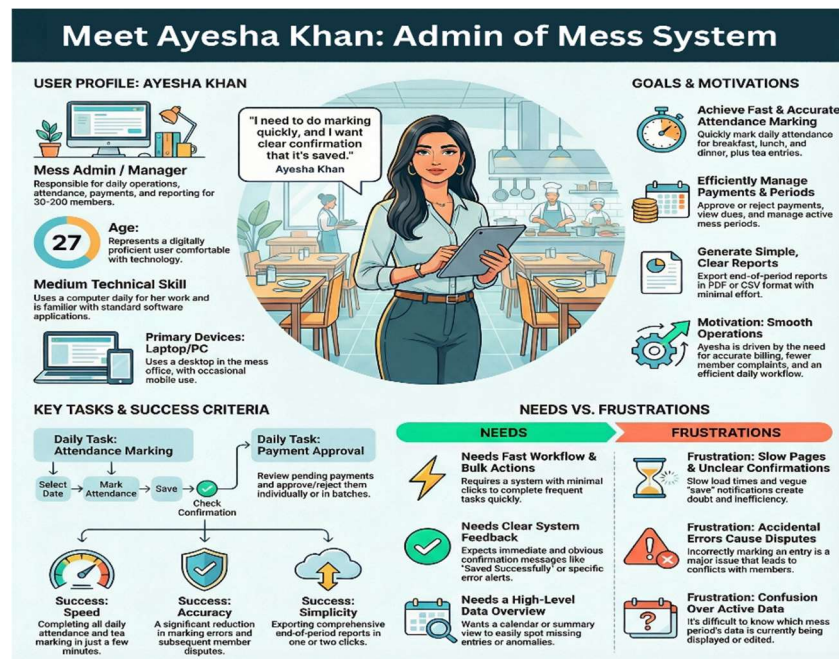
Manual attendance and tea marking often leads to **mistakes**, which causes disputes between Admin and members. Large number of **lists** increase cognitive load and make it **difficult** to quickly find information. Users also face confusion when switching between the current period and past periods, leading to misunderstanding of records.

Usage Scenarios

In daily use, the **admin marks attendance** by selecting the date, updating meal entries, saving the data, and receiving confirmation. **Members** use the system to prevent disputes by **reviewing pending attendance** or tea records and verifying or declining them. At the end of the month, the admin checks payment summaries, reviews outstanding balances, and exports reports for record-keeping.

1d) Personas and User Stories

Admin Persona

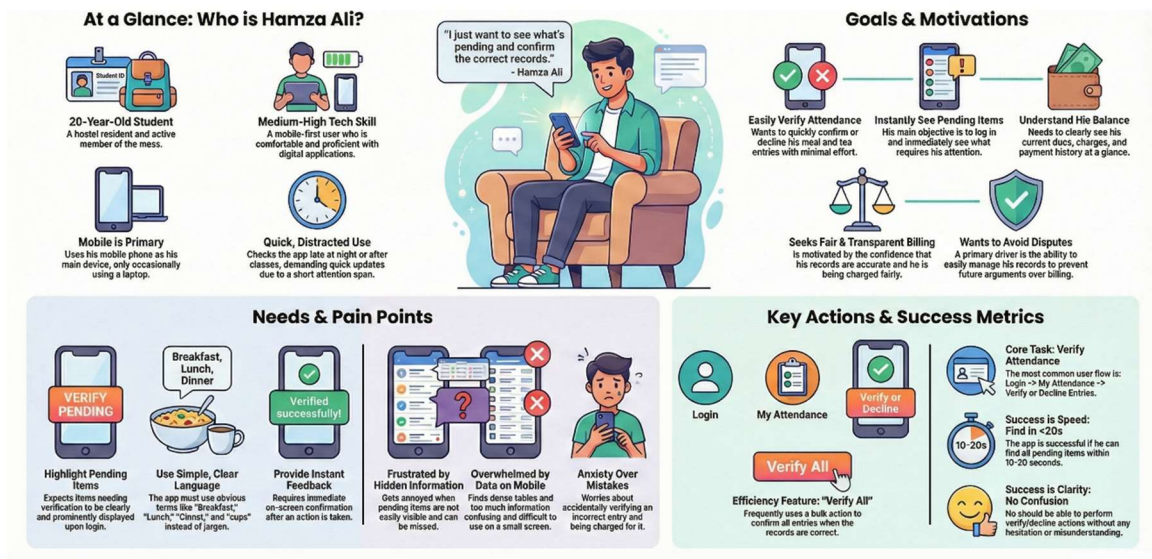


User stories (Admin)

- As an admin, I want to mark attendance for all members for a selected date so that meal charges are accurate.
- As an admin, I want a period calendar view so I can instantly see which days have attendance data.
- As an admin, I want to add dish plans so that I can use and manage them.
- As an admin, I want to add teas of a member so that I can maintain record of member
- As an admin, I want to approve or reject pending payments so that billing remains consistent.
- As an admin, I want to export a period report as PDF/CSV so I can share and archive it.

Member Personas:

Meet Hamza Ali: Member of Mess System



User stories (Member)

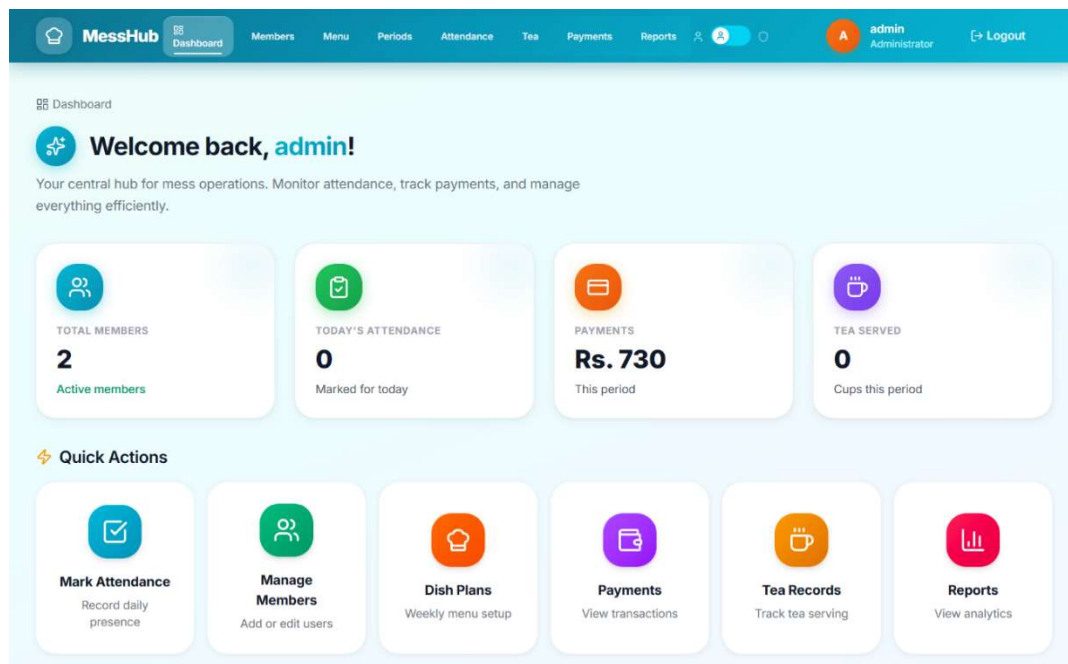
- As a member, I want to see all pending meal verifications so I can confirm or dispute them.
- As a member, I want to view all the weekly dishes plan.
- As a member, I want to verify all pending items quickly when they're correct.
- As a member, I want to see tea cups and charges so I can track my spending.
- As a member, I want to submit payment through cards or other methods

2) Interface Design & Usability Principles

2a) Interface Layout, Navigation Clarity, and Screen Flow

The interface is designed with a clear visual hierarchy so that the most common tasks are easy to find for both Admins and Members. A **sticky top navigation bar** remains visible while scrolling, allowing quick access to main modules such as Attendance, Tea, and Payments. For Admin users, the **dashboard** works as a central control panel that shows **important information** like today's meal count and pending payments as soon as they log in. This reduces the time needed to search for data.

Navigation clarity is improved by using **breadcrumbs** on detailed pages, which help users understand where they are in the system (for example, Dashboard > Attendance > Mark Attendance). The **screen flow** follows a **simple** and logical sequence. In attendance marking, the admin selects a date, marks members for that day, saves the data, and then receives confirmation, which completes the task smoothly.

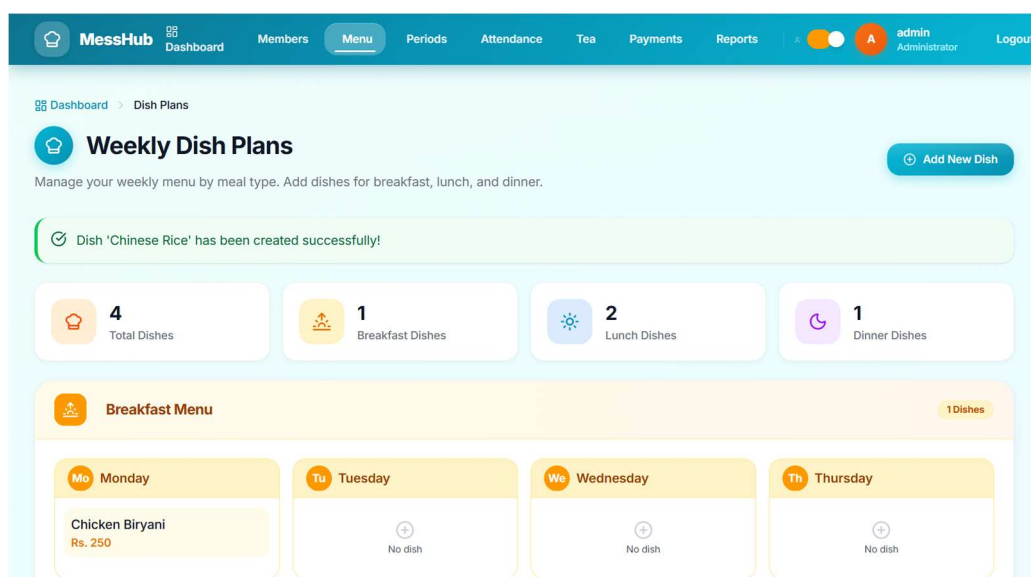


Admin Dashboard with top navigation bar, the breadcrumb trail, and the summary statistics cards.

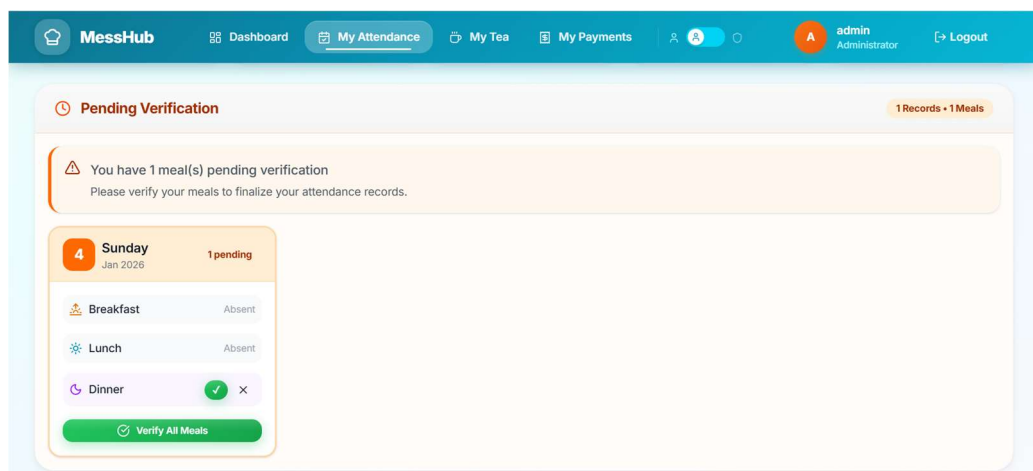
2b) Application of Usability Heuristics

The system applies Nielsen's usability heuristics to make interaction simple and reliable. To maintain visibility of system status, clear **success messages** appear after important actions such as saving attendance or approving payments. The system uses **real-world terms** like Breakfast, Lunch, Mess Period, and currency in Rupees, which makes it easy for users to understand without technical knowledge.

User control is provided by allowing members to **verify** or decline attendance and tea **records** if they are incorrect. Consistency is maintained across the interface by using the same button styles, colors, and layouts in all modules. This consistency helps users learn the system quickly and reduces confusion.



Success message appearing at the top of the screen after adding a dish Menu

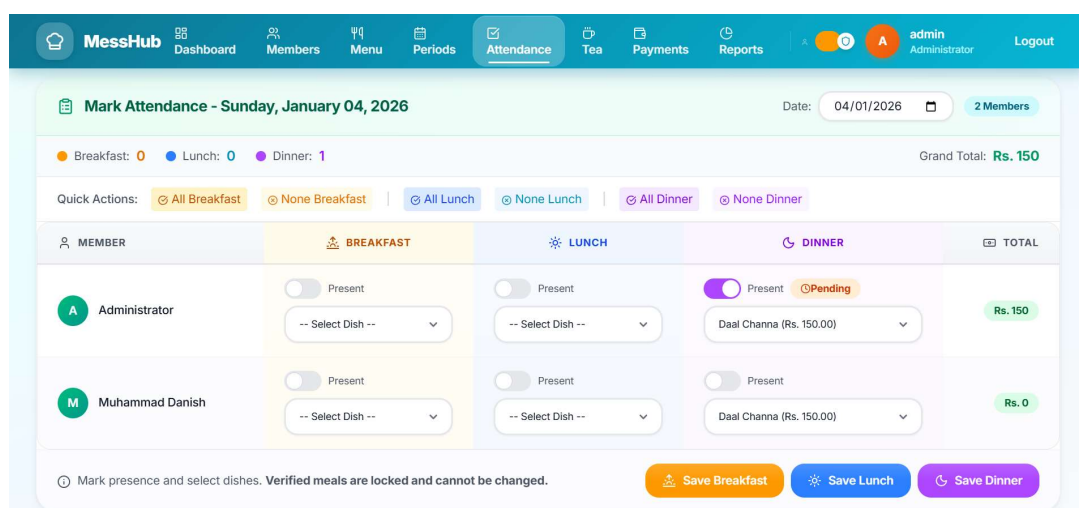


In My Attendance, "Verify" and "Decline" buttons clearly showing to member.

2c) Cognitive & Physical Load Considerations

Cognitive load is reduced by showing **summary cards** on the dashboard instead of large tables at first glance. These cards help users quickly understand the current situation before viewing detailed data. In attendance sheets, **color-coded rows** (such as green for present and grey for absent) allow admins to scan information easily without reading every entry.

Physical load is minimized by using large buttons, checkboxes, and touch-friendly controls, which are especially useful on mobile devices. The responsive design adjusts the layout for smaller screens, preventing horizontal scrolling and making the system easier to use for long periods.

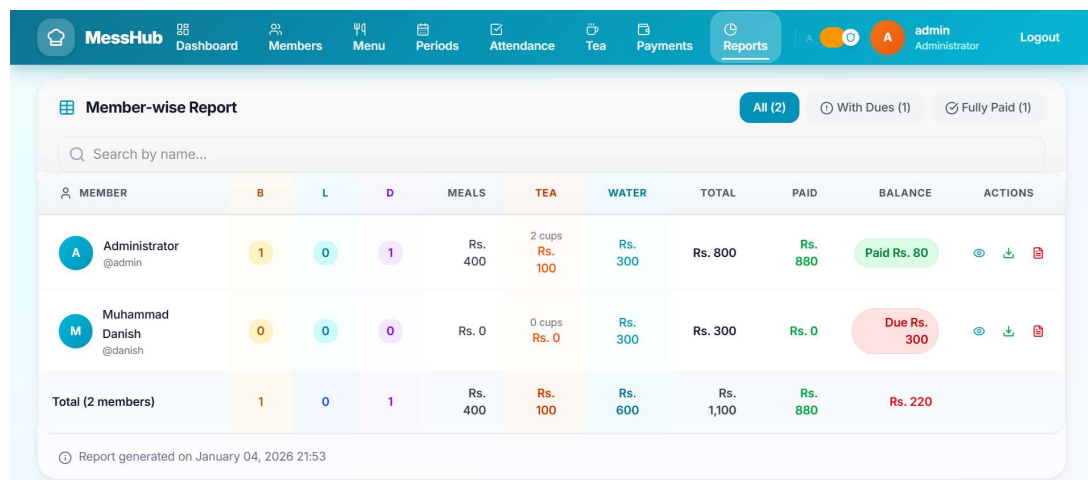


Attendance Marking sheet showing the layout of toggles and rows.

2d) Appropriateness of Color, Typography, Iconography

The system uses **meaningful colors** to communicate status clearly. Green is used for verified or paid records, orange for pending actions, and red for unpaid or error states. This allows users to understand the status of records quickly without reading extra text.

Typography is based on the **Inter font**, which is simple, modern, and easy to read on both desktop and mobile screens. **Icons** are used together with text labels for actions and navigation. This helps users recognize actions faster and improves overall usability.



MEMBER	B	L	D	MEALS	TEA	WATER	TOTAL	PAID	BALANCE	ACTIONS
Administrator @admin	1	0	1	Rs. 400	2 cups Rs. 100	Rs. 300	Rs. 800	Rs. 880	Paid Rs. 80	👁️ ⬇️ 🗑️
Muhammad Danish @danish	0	0	0	Rs. 0	0 cups Rs. 0	Rs. 300	Rs. 300	Rs. 0	Due Rs. 300	👁️ ⬇️ 🗑️
Total (2 members)	1	0	1	Rs. 400	Rs. 100	Rs. 600	Rs. 1,100	Rs. 880	Rs. 220	

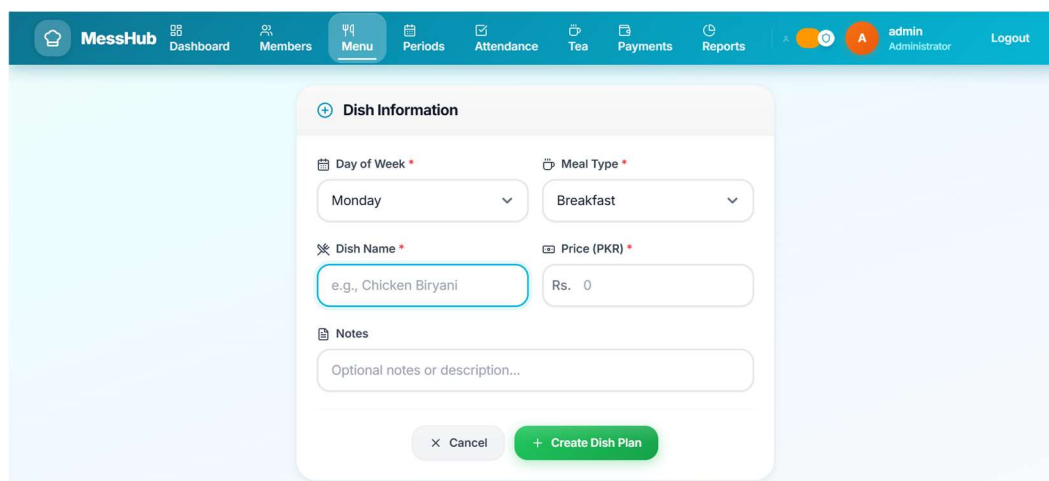
Report generated on January 04, 2026 21:53

Member wise report table showing different colored columns and badges (e.g., Green for 'Paid').

2e) Accessibility Design (contrast, readability, inclusive options)

Accessibility has been considered to ensure the system can be used by a wide range of users. The interface provides good **color contrast** between text and backgrounds, making content readable and visible **focus indicators** show which element is currently selected.

Form fields include proper labels, and **icon-only buttons** use descriptive labels for screen readers. The responsive layout supports zooming and different screen sizes, ensuring that all features remain usable on smaller devices.



MessHub

Dashboard

Members

Menu

Periods

Attendance

Tea

Payments

Reports

admin Administrator

Logout

+

Dish Information

Day of Week *

Monday

Meal Type *

Breakfast

Dish Name *

e.g., Chicken Biryani

Price (PKR) *

Rs. 0

Notes

Optional notes or description...

Cancel

Create Dish Plan

Create Dish form is highlighted with icons and a focus ring on the current input field.

3) Prototype Development (Low-/High-Fidelity)

3a) Low-Fidelity Prototype (Wireframes)

Low-fidelity wireframes were created at the early stage of the project to plan the layout, navigation structure, and task flow of the MessHub system. These wireframes intentionally avoid visual styling such as colors and icons and instead focus on content placement, screen

hierarchy, and usability. This helped validate the workflow for Admin and Member users before moving to implementation.

The wireframes cover all major modules of the system, including attendance marking, tea marking, member verification, payments, and reports.

Account Wireframes:

Login Page

Username

Password

Signin

Forget your Password

Footer

Forget Password Page

Enter Email

Send Link Btn

Login Back

Footer

Admin Wireframes:

Admin Dashboard

Navbar

Welcome Message

Stats

Quick Actions

Recent Activity

Manage Current Period

Footer

Admin Members Page

Navbar

Member Management

Add Member

Members Stats

All Members Table

Informations	View	Edit	Diactivate
Informations	View	Edit	Diactivate
Informations	View	Edit	Diactivate

Footer

Admin Menu Page

Navbar

Weekly Dish PlanAdd Dish

Dishes Stats

Breakfast Menu
Weekly Plans

Lunch Menu
Weekly Plans

Dinner Menu
Weekly Plans

Footer

Admin Mess Period Page

Navbar

Mess PeriodCreate New Period

Period Stats

All Periods Table
Period InformationsEditDelete
Period InformationsEditDelete
Period InformationsEditDelete
Period InformationsEditDelete

Footer

Admin Attendance Page

Navbar

Attendance Stats

Period Attendance CalendarPending Verification

Mark Attendance Section
BreakfastLunchDinner

Footer

Admin Tea Page

Navbar

Attendance Stats

Period Tea CalendarPending Verification

Mark Tea Section
Members with No of Tea Informations

Footer

Admin Payment Page

Navbar

Payment ManagementPending ApprovalAll Record

Select Billing Period

Payment Stats

Member Payment Summary
Member Payment InformationnDetails
Member Payment InformationnDetails
Member Payment InformationnDetails

Footer

Admin Report Page

Navbar

Monthly Record

Select Billing Period

Period StatsExport OptionsEmail All Bills

Member Wise Report
Member Payment InformationViewExport Options
Member Payment InformationViewExport Options
Member Payment InformationViewExport Options

Footer

Member Wireframes:

Member Dashboard Page

Navbar

Welcome Message

My
attendance

View Menus

Payments
Add

Need Help

Footer

Member Attendance Page

Navbar

My Attendance

Attendance Stats

Verify Attendance Section

Attendance History

Footer

Member Tea Page

Navbar

My Tea Records

Tea Stats

Verify Tea Section

Tea Record History

Footer

Member Payment Page

Navbar

My Payment & usage

Submit Payment

Select Period

Charges Breakdown

Daily Usage Detail

Payment History

User Guide

Footer

Member Submit Payment Page

Navbar

Member Name

Outstanding Balance

Payment Details

Period Selection

Amount

Payment Method

Transaction No

Remarks

Approval
Process Details

Cancel

Submit Payment

Footer

Member Menu Page

Navbar

Weekly Menu

Back

Meal Type Counts

Today Menu List

Complete Weekly Schedule

User Guide

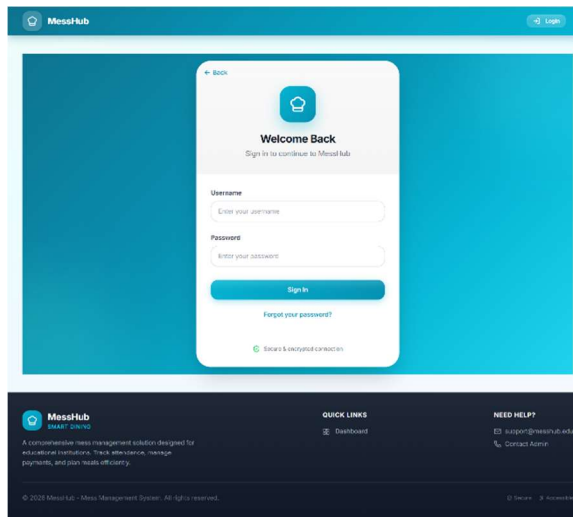
Footer

3a) High-Fidelity Prototype (Final Implemented Screens)

After validating the wireframes, high-fidelity prototypes were developed that closely resemble the final working system. These screens include real data, role-based access, color-coded statuses, confirmation messages, and fully interactive elements. The final prototype follows consistent navigation, meaningful color usage, and accessibility considerations.

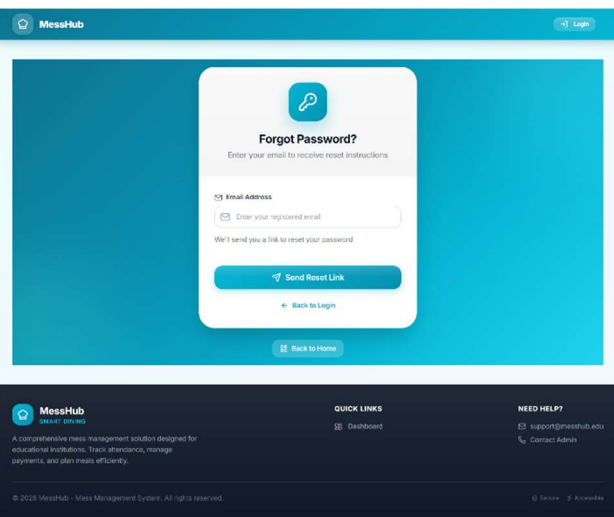
Account Prototypes

Login Page



The Login Page features a blue header with the 'MessHub' logo and a 'Login' button. The main content area has a light blue background with a white login card. The card includes a 'Welcome Back' message, a 'Sign in to continue to MessHub' instruction, and fields for 'Username' and 'Password'. A 'Sign In' button is at the bottom of the card, with a 'Forgot your password?' link below it. A 'Secure & encrypted connection' indicator is at the bottom left of the card.

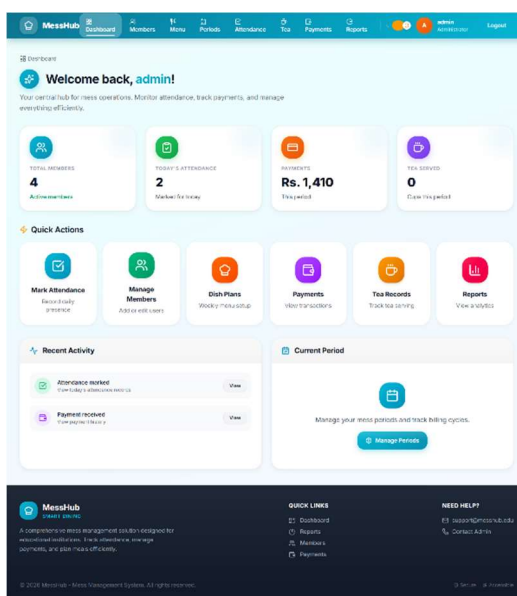
Forgot Password



The Forgot Password page features a blue header with the 'MessHub' logo and a 'Login' button. The main content area has a light blue background with a white card. The card includes a 'Forgot Password?' message, an instruction to 'Enter your email to receive reset instructions', and a field for 'Email Address'. A 'Send Reset Link' button is at the bottom of the card, with a 'Back to Login' link below it. A 'Back to Home' button is at the bottom right of the card.

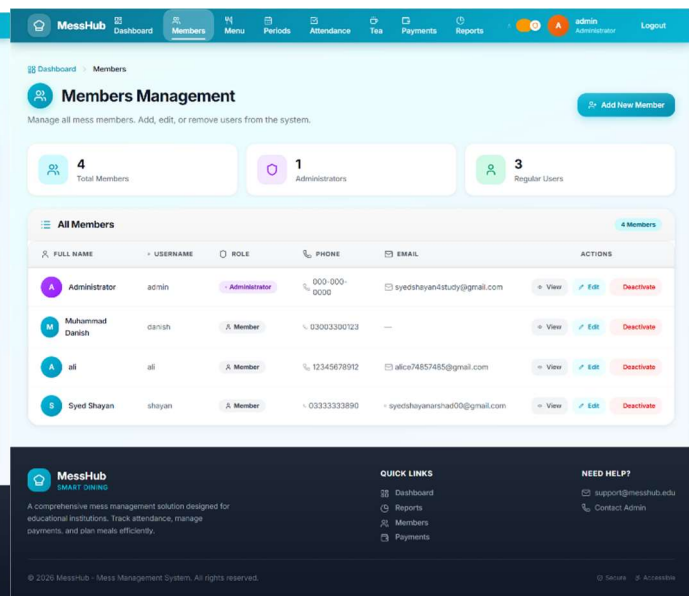
Admin Side

Dashboard



The Admin Dashboard features a blue header with the 'MessHub' logo and a 'Logout' button. The main content area has a light blue background. It includes a 'Welcome back, admin!' message, a 'Your central hub for mess operations' description, and four summary cards: 'Total Members: 4', 'Today's Attendance: 2', 'Amount: Rs. 1,410', and 'No debts: 0'. Below these are 'Quick Actions' for 'Mark Attendance', 'Manage Members', 'Dish Plans', 'Payments', 'Tax Records', and 'Reports'. A 'Recent Activity' section shows 'Attendance marked' and 'Payment received'. A 'Current Period' section shows 'Manage your mess periods and track billing cycles' with a 'Manage Periods' button.

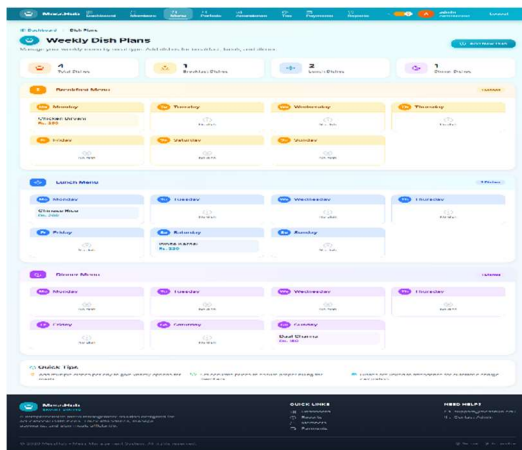
Members Page



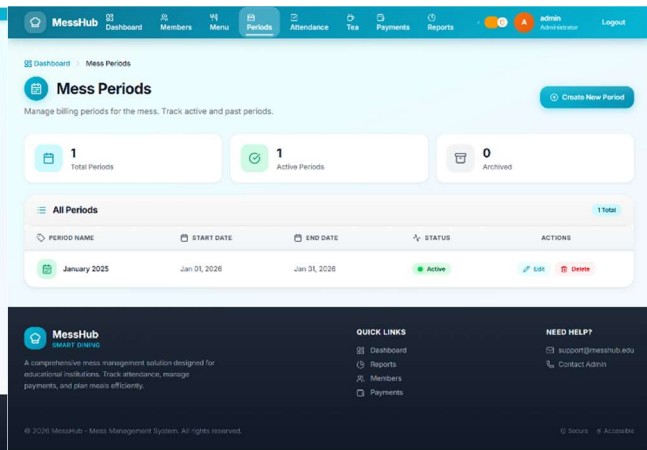
The Members Page features a blue header with the 'MessHub' logo and a 'Logout' button. The main content area has a light blue background. It includes a 'Members Management' section with a 'Manage all mess members. Add, edit, or remove users from the system.' description and an 'Add New Member' button. Below this are three summary cards: '4 Total Members', '1 Administrators', and '3 Regular Users'. A table lists all members with columns for 'Full Name', 'Username', 'Role', 'Phone', 'Email', and 'Actions'. The table contains four rows of data.

Full Name	Username	Role	Phone	Email	Actions
Administrator	admin	Administrator	000-000-0000	syedshayanashat00@gmail.com	View Edit Deactivate
Muhammad Danish	danish	Member	03003300123	—	View Edit Deactivate
ali	ali	Member	12345678912	alice74857485@gmail.com	View Edit Deactivate
Syed Shayan	shayan	Member	03333333890	syedshayanashat00@gmail.com	View Edit Deactivate

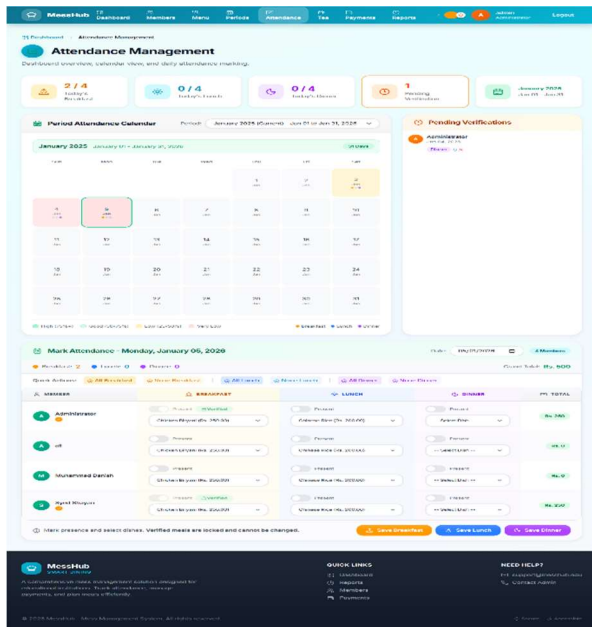
Menu Page



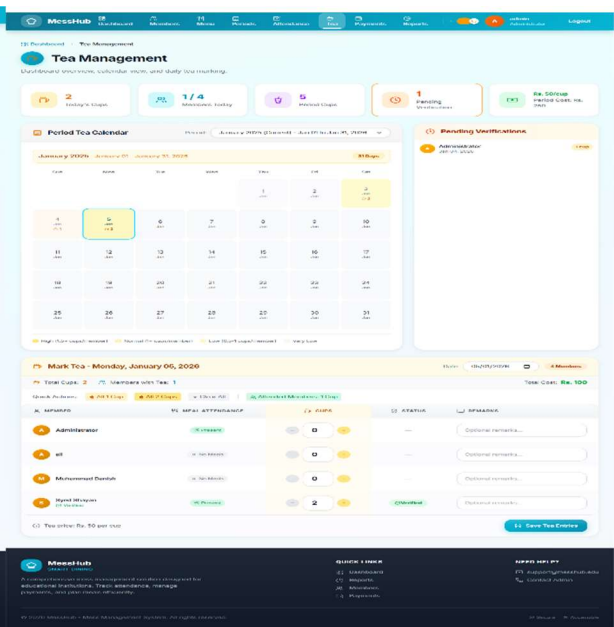
Periods Page



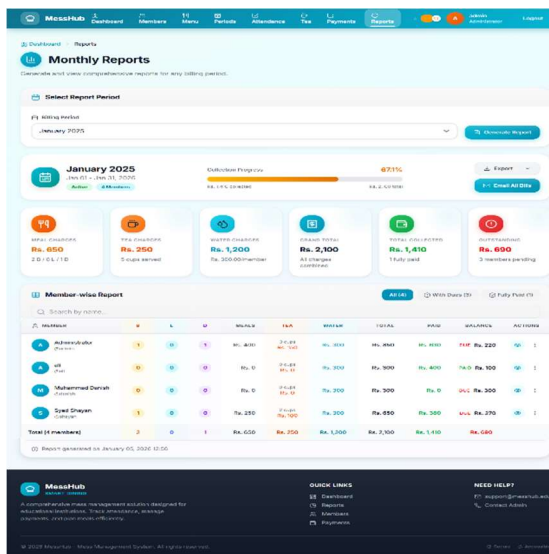
Attendance Page



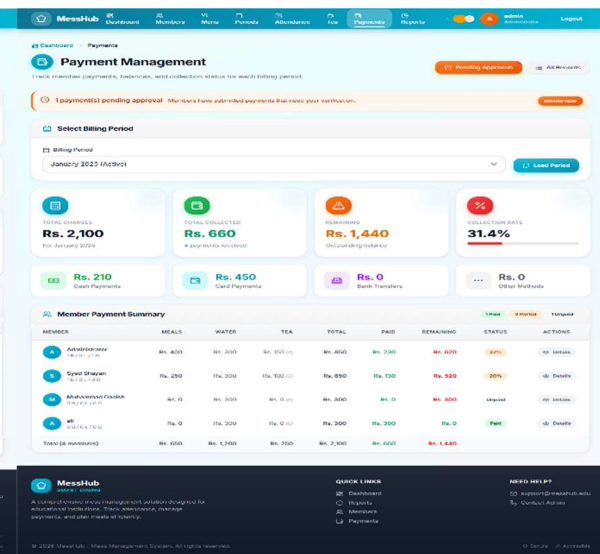
Tea Page



Payment Page

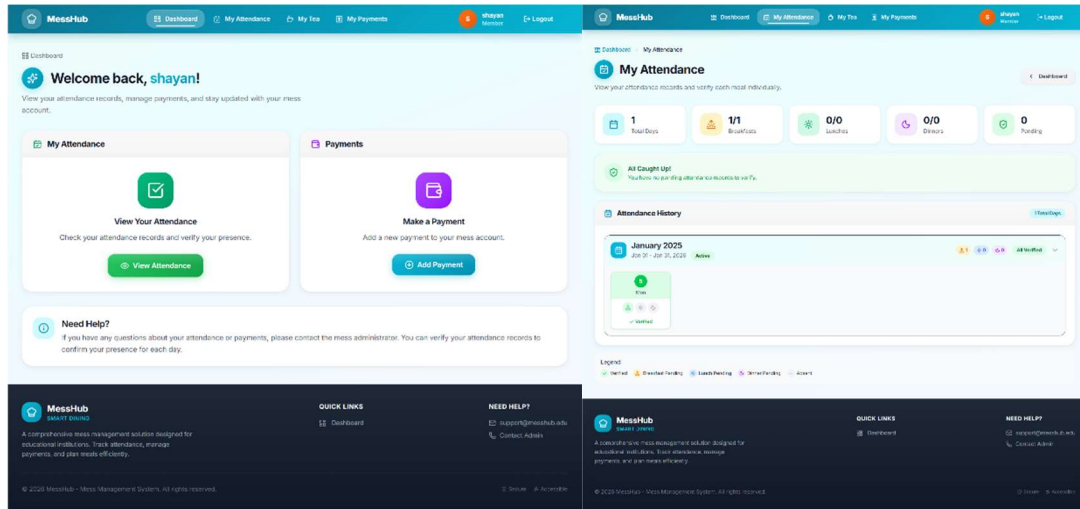


Report Page

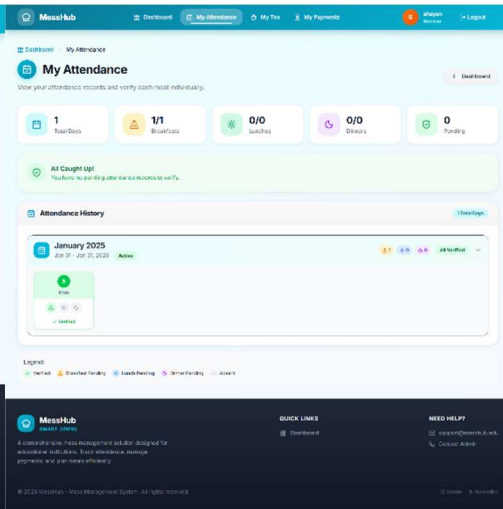


Member Pages:

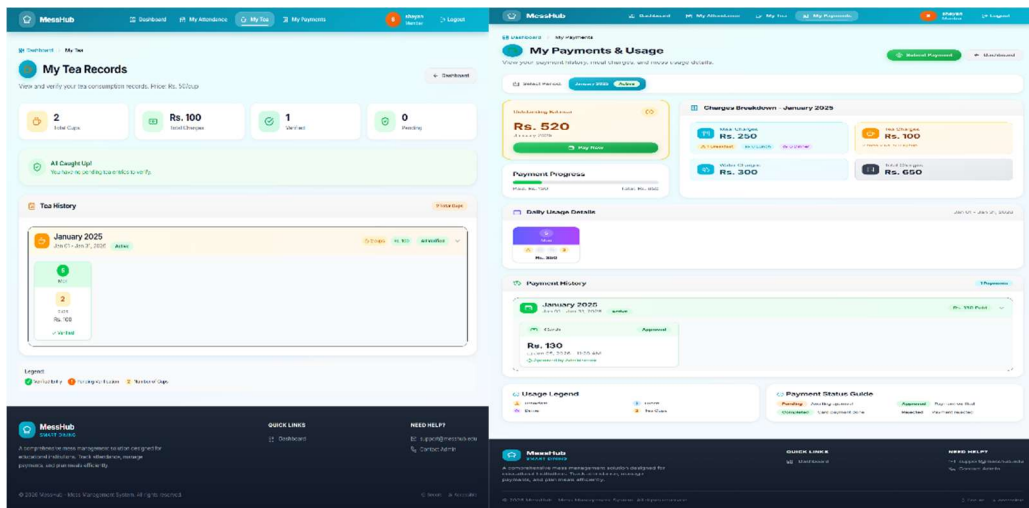
Dashboard Page



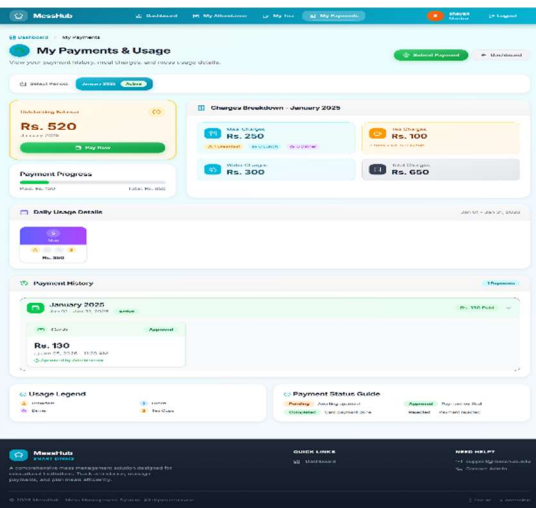
Attendance Page



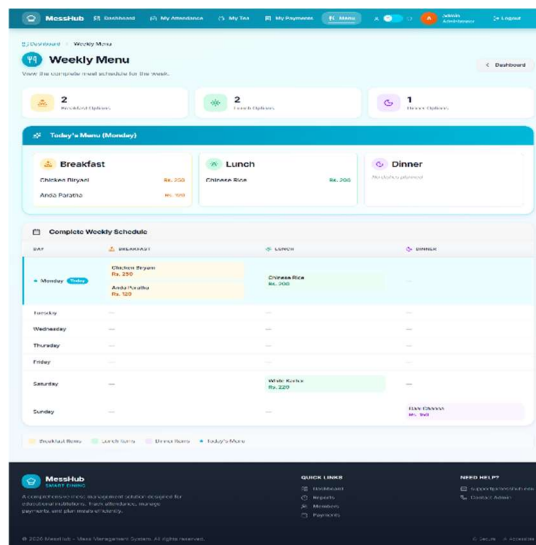
Tea Page



Payment Page



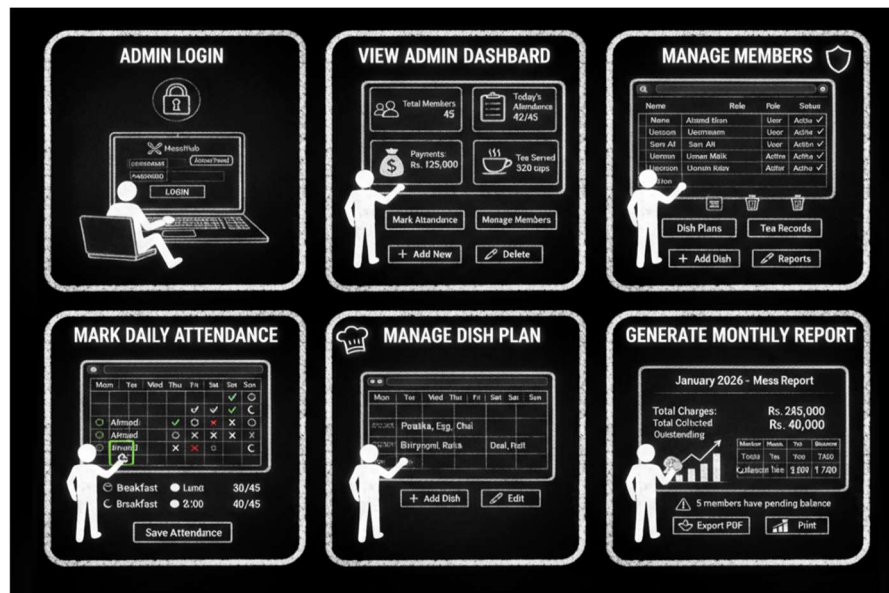
Menu Page



3b) Coverage of Core Features

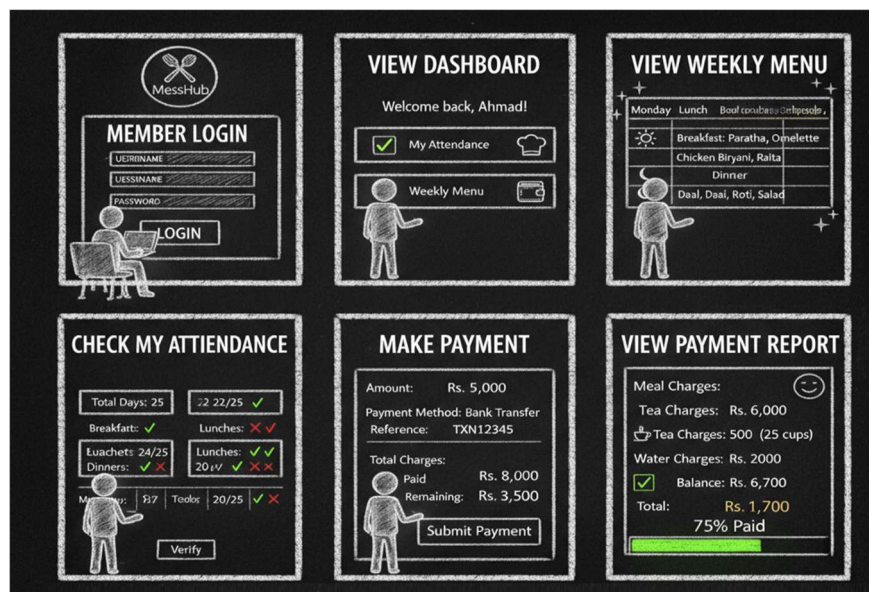
The developed prototype of MessHub successfully covers all core features defined in the system requirements. For **Admin** users, the prototype **includes** login and account recovery, dashboard overview, member management, menu and dish plan management and all other main features. These are accessible through the main navigation and follow a consistent layout.

Admin Story Board



For **Member** users, the prototype **supports** dashboard access, viewing and verification of attendance records, and all main features. These features provide transparency and allow members to actively participate in verifying their records.

Member Story Board



Overall, the prototype reflects all **functional requirements** of the system and ensures that no critical workflow is missing.

3c) Mapping Between User Actions and System Responses

The prototype shows a strong connection between user actions and system responses by providing immediate and visible feedback. Actions such as **marking attendance** or tea entries display clear **success messages**, while **payment approvals** or rejections update the status with **confirmation**.

For **members**, verifying or declining attendance and tea records instantly updates **status badges**, and submitted payments are shown as pending until admin approval. **Errors** such as missing inputs or invalid actions are communicated through **clear messages**, helping reduce confusion and build user trust.

3d) Demonstration of Complete Task Flows

The prototypes explain complete end-to-end task flows for all major system operations. Each task follows a logical sequence that allows users to filter members to complete their goals without unnecessary steps. For example, the daily attendance marking flow for the admin works as follows:

1. Admin logs into the system
2. Admin navigates to the Attendance page and select specific date
3. Breakfast, Lunch, and Dinner attendance is marked for members and save it.
4. The system confirms successful saving of data

The screenshot displays the 'Mark Attendance' interface for Monday, January 05, 2026. At the top, it shows the date and the number of members (4). Below this, there are quick actions for Breakfast, Lunch, and Dinner. The main table lists members and their attendance status for each meal. The 'Lunch' column is highlighted with a green box, and the 'Save Lunch' button is also highlighted. A success message at the bottom states 'Lunch attendance saved successfully!'. The bottom navigation bar includes options like Dashboard, Members, Menu, Periods, Attendance, Tea, Payments, Reports, and a user profile for 'admin Administrator'.

MEMBER	BREAKFAST	LUNCH	DINNER	TOTAL
Administrator	Present (Verified) Chicken Biryani (Rs. 250.00)	Present Chinese Rice (Rs. 200.00)	Present -- Select Dish --	Rs. 250
ali	Present Chicken Biryani (Rs. 250.00)	Present Chinese Rice (Rs. 200.00)	Present -- Select Dish --	Rs. 200
Muhammad Danish	Present Chicken Biryani (Rs. 250.00)	Present Chinese Rice (Rs. 200.00)	Present -- Select Dish --	Rs. 200
Syed Shayam	Present (Verified) Chicken Biryani (Rs. 250.00)	Present Chinese Rice (Rs. 200.00)	Present -- Select Dish --	Rs. 250

3e) Realism, Intuitiveness, and Implementation Quality

The **high-fidelity prototype** of MessHub is **realistic** and easy to use, with clear navigation, familiar terms, readable text, and effective color coding. It **supports** quick daily tasks for

Admin users while giving Members **clear visibility** and control. Overall, the design follows standard usability principles and is suitable for real-world mess management.

4) Evaluation & Testing

4a) Selection of Evaluation Methods

To evaluate the MessHub system, we used a combination of **Heuristic Evaluation** and **User-Based Testing** during the development phase of the project. This allowed us to identify interface-level issues as well as problems faced by users while performing real tasks. Heuristic Evaluation **focused on usability** principles such as consistency, error prevention, and user control. **User-Based Testing** involved **practical tasks** like marking attendance, verifying meals, and making payments. This mixed approach helped ensure that the system is both usable and effective in real-world scenarios.

4b) Reporting Findings with Clarity

The evaluation process revealed several usability issues and pain points. One major issue was related to **attendance marking**, where selecting only lunch attendance automatically marked breakfast and dinner as well, leading to **incorrect records**. Another **usability issue** was the absence of a **Forgot Password** feature, which made account recovery difficult for users. **Payment-related problems** were also identified, such as missing minimum payment constraints in Stripe and **unclear feedback** when admins approved non-Stripe payments.

User feedback highlighted **confusion** in attendance and tea **verification**, as users were initially only able to verify records but not decline incorrect entries. **Exporting reports** was also **problematic**, as files did not download correctly. These **findings** were clearly documented and categorized to guide **improvements**.

Module	Issues Identified	Impact on User	Severity	Evaluation Method
Attendance Marking	Marking only lunch automatically marked breakfast and dinner	Incorrect attendance and billing errors	High	Usability Testing
Verification	Users could only verify records, no option to decline incorrect entries	Loss of trust	High	Usability Testing
Admin Controls	Admin could not cancel meals marked by mistake	Data errors difficult to fix	Medium	Usability Testing
Authentication	No “Forgot Password” option initially available	Users could not recover accounts	High	Heuristic Review
Stripe Payment	No minimum payment constraint applied	Unrealistic payments accepted	Medium	Heuristic Review
Payment Approval	No confirmation feedback when admin approved manual payments	Confusion about action success	Low	Usability Testing
Reports	Export files did not download correctly	Reports unusable for record keeping	Medium	Usability Testing
Member Management	Members were permanently deleted instead of deactivated	Risk of data loss	High	Heuristic Review

4c) Comparison of Methods Used (Expert Review vs. User Testing)

Different insights were gained from controlled testing and real usage scenarios. During **lab-style testing and heuristic review**, design and rule-based issues were identified, such as missing **constraints on payment** values and lack of **safety features** like member deactivation instead of deletion. These issues were easier to detect through systematic inspection.

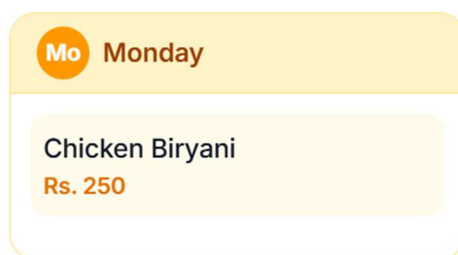
In contrast, **real environment usage** revealed practical workflow problems. For example, users were confused when **exporting reports** or when approving manual payments due to missing confirmation feedback. Real usage also exposed trust-related issues, such as the need for decline options and **admin-level cancellation** of mistakenly marked meals. This comparison shows that while lab testing helps in catching design flaws early, real environment testing is essential for understanding actual user behavior.

4d) Evidence of Improvement (Iteration)

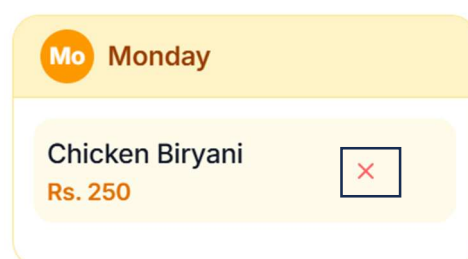
Based on feedback from evaluation, several **improvements** were implemented. **Attendance marking** was corrected so that meals are handled independently. A “Forgot Password” feature with **email support** was added. Minimum payment **limits** were applied to Stripe payments, and export functionality was fixed to ensure reports download correctly.

Additional improvements included introducing member **deactivation** instead of deletion, adding **decline** options for attendance and tea **records**, allowing admins to cancel mistakenly marked meals, and improving feedback during manual payments. These changes clearly show an iterative design process where user feedback directly influenced system improvements.

Before

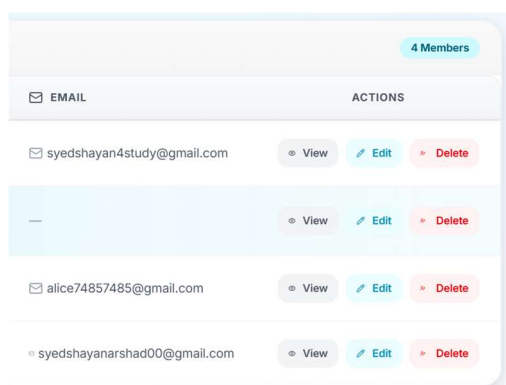


After

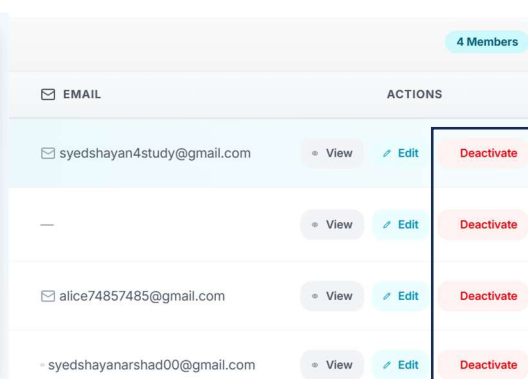


Add remove option to delete meal in dish plan.

Before



After



Change User from being delete to deactivate.